Year 11 Mathematics

Number 9

6.2 Percentage Increase and Decrease

Percentage Increase and Decrease

STARTER · 5 MIN

Date: _____ **Title:** Percentage Increase and Decrease

LEARNING OBJECTIVES

- Increase and decrease by a percentage
- Use multipliers for increase and decrease
- Find a percentage change, profit or loss

STARTER *Answer in your book. No calculators.*

1. Work out 1 + 0.2
2. Work out 1 − 0.15
3. Find 10% of 450
4. Write 105% as a decimal

Starter — Answers

STARTER

Tick or correct in green pen

1. **1.2**
2. **0.85**
3. **45**
4. **1.05**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can increase and decrease by a percentage	☐
SILVER	I can use multipliers for increase and decrease	☐
GOLD	I can find a percentage change, profit or loss	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Increase and Decrease

INTRODUCTION · 3 MIN

Find the percentage and add or subtract it, or use one multiplier.

Increase

Find the percentage, then add it.

£50 + 10% = £55

Multiplier up

100% + the increase.

Increase 20% → × 1.2

Multiplier down

100% − the decrease.

Decrease 15% → × 0.85

Percentage change

Change ÷ original × 100.

20 → 25 is a 25% increase

Examples 1 and 2 — Add or Subtract the Percentage

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Example 1

Increase £80 by 15%.

Example 2

Decrease 250 kg by 30%.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

15% of 80 = 12

80 + 12

£92

Tip: 10% + 5%.

Example 2

30% of 250 = 75

250 − 75

175 kg

Tip: subtract for a decrease.

Task 1

PRACTICE · 9 MIN

BRONZE *No calculators*

1. Increase £60 by 10%
2. Decrease 200 by 25%
3. Increase 40 kg by 5%
4. Decrease £120 by 35%
5. Increase 300 m by 12%

★ Decrease £18 by 20%

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $60 + 6 =$ **£66**
2. $200 - 50 =$ **150**
3. $40 + 2 =$ **42 kg**
4. $120 - 42 =$ **£78**
5. $300 + 36 =$ **336 m**

★ $18 - 3.60 =$ **£14.40**

Examples 3 and 4 — Using Multipliers

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint increase: 1 + rate decrease: 1 − rate

Example 3

Increase £340 by 18%.

Example 4

Decrease 85 kg by 6%.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiplier = 1.18

1.18 × 340

£401.20

Tip: 100% + 18% = 118%.

Example 4

Multiplier = 0.94

0.94 × 85

79.9 kg

Tip: 100% − 6% = 94%.

Task 2

PRACTICE · 9 MIN

SILVER *Use a multiplier*

Hint increase: 1 + rate decrease: 1 − rate

1. Increase £250 by 7%
2. Decrease £64 by 15%
3. Increase 1200 by 3.5%
4. Decrease 48 m by 12.5%
5. A £15 000 car loses 22% of its value. Find its new value.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $1.07 \times 250 = \mathbf{£267.50}$
2. $0.85 \times 64 = \mathbf{£54.40}$
3. $1.035 \times 1200 = \mathbf{1242}$
4. $0.875 \times 48 = \mathbf{42\ m}$
5. $0.78 \times 15\,000 = \mathbf{£11\,700}$

Examples 5 and 6 — Percentage Change

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Example 5

(2 marks)

A price rises from £40 to £46. Find the percentage increase.

Example 6

(3 marks)

A shop buys a jacket for £60 and sells it for £84. Find the percentage profit.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Change = 46 − 40

= 6

6 ÷ 40 × 100

15%

Tip: always divide by the original.

Example 6

Profit = 84 − 60

= £24

24 ÷ 60 × 100

40%

Tip: profit ÷ cost price × 100.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

1. A population rises from 2500 to 2800. Find the percentage increase. (2 marks)

2. A phone falls from £450 to £378. Find the percentage decrease. (2 marks)

3. Bought for £120, sold for £150. Find the percentage profit. (2 marks)

4. Bought for £80, sold for £68. Find the percentage loss. (2 marks)

5. £50 is increased by 20%, then the new price is decreased by 20%. Find the final price. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $300 \div 2500 \times 100 = \mathbf{12\%}$

2. $72 \div 450 \times 100 = \mathbf{16\%}$

3. $30 \div 120 \times 100 = \mathbf{25\%}$

4. $12 \div 80 \times 100 = \mathbf{15\%}$

5. $1.2 \times 50 = 60, \quad 0.8 \times 60 = \mathbf{£48}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Increase £70 by 10%

SILVER Decrease £90 by 15% using a multiplier

GOLD A value goes from 50 to 65. Find the percentage increase.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $70 + 7 = \mathbf{£77}$

SILVER $0.85 \times 90 = \mathbf{£76.50}$

GOLD $15 \div 50 \times 100 = \mathbf{30\%}$

Exam Practice — Increase and Decrease

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: answer on the line, show all your working

Question 1 (2 marks)

Increase £140 by 15%

Do not use a calculator.

..... £

Question 2 (2 marks)

A coat costs £85.

In a sale the price is reduced by 20%.

Work out the sale price.

..... £

Exam Practice — Using Multipliers

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint increase: 1 + rate decrease: 1 − rate

Question 3

(2 marks)

Increase £360 by 4.5%

..... £

Question 4

(2 marks)

A car is worth £18 500.

Its value falls by 13%.

Work out its new value.

..... £

Exam Practice — Percentage Change

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Question 5

(3 marks)

A shop buys a lamp for £36.

It sells the lamp for £45.

Work out the percentage profit.

..... %

Question 6

(3 marks)

A price is increased by 10%.

The new price is then decreased by 10%.

Is the final price the same as the original?

Give a reason for your answer.

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

15% of 140 = 14 + 7 = 21 — M1

140 + 21 = £161 — A1

Tip: 10% + 5%.

Question 2

20% of 85 = 17 — M1

85 − 17 = £68 — A1

Tip: subtract for a decrease.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

1.045 × 360 — M1

= £376.20 — A1

Tip: 100% + 4.5% = 104.5%.

Question 4

0.87 × 18 500 — M1

= £16 095 — A1

Tip: 100% − 13% = 87%.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

profit = 45 − 36 = £9 — M1

$\frac{9}{36} \times 100$ — M1

= 25% — A1

Tip: divide by the cost price.

Question 6

1.1 × 0.9 = 0.99 — M1

final price = 99% of original — M1

No, it is 1% lower — A1

Tip: the 10% decrease is of a bigger amount.